

Template Title

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Question I

0.1 Gell-Man as a Basis of $\mathfrak{su}(3)$

Definition 0.1.1 $\mathfrak{su}(N)$. $\mathfrak{su}(N)$

$$\mathfrak{su}(n) = \left\{ X \in M_{n \times n}(\mathbb{C}) \mid X^\dagger = -X, \text{Tr } X = 0, \det X = 1 \right\} \quad (1)$$

The dimension of $\mathfrak{su}(n)$ is $\dim \mathfrak{su}(3) = n^2 - 1$

Definition 0.1.2. Let X_i be a basis of a lie algebra. Then, since a lie algebra is a vector space, the Lie bracket is a superposition of the basis

$$[X_i, X_j] = f_{ij}^k X_k \quad (2)$$

where f_{ij}^k are the structure constants.

Definition 0.1.3 Generators of $\mathfrak{su}(3)$.

We define the Hermitian generators $\lambda^a \equiv \frac{\lambda^a}{2}$ for $a = 1, \dots, 8$

Remark 0.1.4.

The lie algebra $\mathfrak{su}(3)$ is the set of 3×3 matrices $\{X : X^\dagger = -X, \text{Tr } X = 0\}$. The dimension is $3^2 - 1 = 8$. The lie algebra $\mathfrak{su}(3)$ is the set of antihermitian traceless 3×3 matrices. We can denote the set of antihermitian matrices by $\mathcal{A}$ and this set forms a real vector space since it is closed under real multiplication and matrix addition (not complex since it isn't closed under complex multiplication). Likewise, the set of hermitian matrices $\mathcal{H}$ also forms a real vector space.

Definition 0.1.5 Real-Linear Bijection. We define the real-linear bijection $\varphi : X \rightarrow iX$ between traceless Hermitian matrices and traceless antiHermitian ones.

If $X \in \mathcal{H}$ then $iX \in \mathcal{A}$, thus φ is an honest map $\varphi : \mathcal{H} \rightarrow \mathcal{A}$.

Let $X, Y \in \mathcal{H}, \lambda, \mu \in \mathbb{R}$

$$\varphi(\lambda X + \mu Y) = i(\lambda X + \mu Y) = \lambda(iX) + \mu(iY) = \lambda\varphi(X) + \mu\varphi(Y) \quad (3)$$

Now if we let $Y \in \mathcal{A}$ The inverse is give by $\varphi^{-1}(Y) = -iY$. $\varphi^{-1}(\varphi(X)) = -i(iX) = X$ and $\varphi(\varphi^{-1}(Y)) = \varphi(-iY) = i(-iY) = Y$. Since there exists an explicit two sided inverse we have a bijection.

Remark 0.1.6.

Since a real-linear bijection sends a real basis of $\mathcal{H}$ to a real basis of $\mathcal{A}$: if $\{\lambda^a\}$ spans $\mathcal{H}$ and is linearly independent then $\varphi(\{\lambda^a\})$ forms a basis for $\mathcal{A}$. Thus we can consider the whether if $\{\lambda^a\}$ is a basis for $\mathcal{H}$ then $\varphi(\{\lambda^a\}) = i\{\lambda^a\}$ is a basis for $\mathcal{A}$.

Directly inspecting each of the λ^a we can see that $\lambda^{a\dagger} = \lambda^a$. Again by inspection we have that they are traceless, i.e. sum of the diagonal is zero. Thus λ^a is a set of traceless Hermitian matrices.

Now we consider linear independence. We require

$$\sum_{a=1}^8 c_a \lambda^a = 0.$$

Thus for a given λ^a we can look at the components ij and form a system of equations.

$$\begin{aligned}
(1,1) \quad c_3 + \frac{c_8}{\sqrt{3}} &= 0 \\
(1,2) \quad -ic_2 &= 0 \\
(1,3) \quad c_4 - ic_5 &= 0 \\
(2,1) \quad c_1 + ic_2 &= 0 \\
(2,2) \quad -c_3 + \frac{c_8}{\sqrt{3}} &= 0 \\
(2,3) \quad c_6 - ic_7 &= 0 \\
(3,1) \quad c_4 + ic_5 &= 0 \\
(3,2) \quad c_6 + ic_7 &= 0 \\
(3,3) \quad -\frac{2c_8}{\sqrt{3}} &= 0.
\end{aligned}$$

$$\begin{aligned}
(1,2), (2,1) &\implies c_1 = c_2 = 0 \\
(1,3), (3,1) &\implies c_4 = c_5 = 0 \\
(2,3), (3,2) &\implies c_6 = c_7 = 0 \\
(1,1), (2,2) &\implies c_3 = c_8 = 0.
\end{aligned}$$

We have eight linearly independent traceless Hermitian matrices in an 8dimensional real vector space necessarily span it XXX so $\{\lambda^a\}$ is a basis for $\mathcal{H}$ and thus $\{iT^a\} = \{i\frac{\lambda^a}{2}\}$ forms a basis for $\mathfrak{su}(3)$.

0.2 Structure constants

$$[T^a, T^b] = if^{abc}T^c \quad (4)$$

Given the normal trace normalisation XXX

$$\text{Tr}(\lambda^a \lambda^b) = 2\delta^{ab} \quad (5)$$

we have

$$\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab} \quad (6)$$

If we multiply (4) by T^d and then take the trace

$$\text{Tr}([T^a, T^b] T^d) = if^{abc} \text{Tr}(T^c T^d) = if^{abc} \frac{1}{2}\delta^{cd} \quad (7)$$

$$\text{Tr}([T^a, T^b] T^d) = if^{abd} \frac{1}{2} \quad (8)$$

Thus we have

$$f^{abd} = -2i \text{Tr}([T^a, T^b] T^d) = -\frac{i}{4} \text{Tr}([\lambda^a, \lambda^b] \lambda^d) \quad (9)$$

1 Question II

We need to show that $S^{\rho\sigma} = \frac{1}{4} [\gamma^\rho, \gamma^\sigma]$ satisfies the Lorentz algebra as long as the γ^μ matrices satisfy the Clifford algebra.

Definition 1.0.1 Lorentz Algebra.

The Lorentz algebra is defined as

$$[S^{\kappa\lambda}, S^{\mu\nu}] = \eta^{\lambda\mu} S^{\kappa\nu} - \eta^{\kappa\mu} S^{\lambda\nu} - \eta^{\lambda\nu} S^{\kappa\mu} + \eta^{\kappa\nu} S^{\lambda\mu}. \quad (1.1)$$

XXX check factor of i

Definition 1.0.2 Clifford Algebra.

The Clifford algebra is defined as

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \mathbb{I}. \quad (1.2)$$

Proof.

We have

$$[S^{\kappa\lambda}, \gamma^\mu] = \left[\frac{1}{4} [\gamma^\kappa, \gamma^\lambda], \gamma^\mu \right] \quad (1.3)$$

$$\begin{aligned} [S^{\kappa\lambda}, \gamma^\mu] &= \frac{1}{4} (\gamma^\kappa \gamma^\lambda \gamma^\mu - \gamma^\lambda \gamma^\kappa \gamma^\mu - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa) \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= -\gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\kappa \gamma^\lambda \gamma^\mu + \gamma^\lambda \gamma^\mu \gamma^\kappa - \gamma^\lambda \gamma^\mu \gamma^\kappa - \gamma^\lambda \gamma^\kappa \gamma^\mu - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= -\gamma^\kappa \gamma^\mu \gamma^\lambda + 2\gamma^\kappa \eta^{\lambda\mu} + \gamma^\lambda \gamma^\mu \gamma^\kappa - 2\gamma^\lambda \eta^{\kappa\mu} - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= +\gamma^\mu \gamma^\kappa \gamma^\lambda - \gamma^\mu \gamma^\kappa \gamma^\lambda - \gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa - \gamma^\mu \gamma^\lambda \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} + \gamma^\lambda \gamma^\mu \gamma^\kappa - 2\gamma^\lambda \eta^{\kappa\mu} \\ &\quad - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= +\gamma^\mu \gamma^\kappa \gamma^\lambda - 2\eta^{\kappa\mu} \gamma^\lambda + 2\eta^{\lambda\mu} \gamma^\kappa - \gamma^\mu \gamma^\lambda \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} - 2\gamma^\lambda \eta^{\kappa\mu} - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= -2\eta^{\kappa\mu} \gamma^\lambda + 2\eta^{\lambda\mu} \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} - 2\gamma^\lambda \eta^{\kappa\mu} \\ 4 [S^{\kappa\lambda}, \gamma^\mu] &= +4\eta^{\lambda\mu} \gamma^\kappa - 4\eta^{\kappa\mu} \gamma^\lambda \end{aligned}$$

We get the result

$$[S^{\kappa\lambda}, \gamma^\mu] = -\eta^{\lambda\mu} \gamma^\kappa + \eta^{\kappa\mu} \gamma^\lambda \quad (1.4)$$

Further, we want to show that

$$[S^{\kappa\lambda}, S^{\mu\nu}] = \eta^{\lambda\nu} S^{\kappa\mu} - \eta^{\lambda\mu} S^{\kappa\nu} + \eta^{\kappa\mu} S^{\lambda\nu} - \eta^{\kappa\nu} S^{\lambda\mu} \quad (1.5)$$

$$\begin{aligned}
[S^{\kappa\lambda}, S^{\mu\nu}] &= \left[S^{\kappa\lambda}, \frac{1}{4} [\gamma^\mu, \gamma^\nu] \right] \\
4 [S^{\kappa\lambda}, S^{\mu\nu}] &= [S^{\kappa\lambda}, \gamma^\mu] \gamma^\nu + \gamma^\mu [S^{\kappa\lambda}, \gamma^\nu] - [S^{\kappa\lambda}, \gamma^\nu] \gamma^\mu - \gamma^\nu [S^{\kappa\lambda}, \gamma^\mu] \\
4 [S^{\kappa\lambda}, S^{\mu\nu}] &= [[S^{\kappa\lambda}, \gamma^\mu], \gamma^\nu] + [\gamma^\mu, [S^{\kappa\lambda}, \gamma^\nu]] \\
4 [S^{\kappa\lambda}, S^{\mu\nu}] &= -\eta^{\lambda\mu} [\gamma^\kappa, \gamma^\nu] + \eta^{\kappa\mu} [\gamma^\lambda, \gamma^\nu] - \eta^{\lambda\mu} [\gamma^\kappa, \gamma^\nu] + \eta^{\kappa\mu} [\gamma^\nu, \gamma^\lambda] \\
[S^{\kappa\lambda}, S^{\mu\nu}] &= -\eta^{\lambda\mu} S^{\kappa\nu} + \eta^{\kappa\mu} S^{\lambda\nu} - \eta^{\lambda\mu} S^{\kappa\nu} + \eta^{\kappa\mu} S^{\nu\lambda}.
\end{aligned}$$

□

Question III

1.1 Part (A)

Given $\Omega_{12} = -\Omega_{21} = i\theta$ and all other terms vanish.

$$\Lambda = e^{-\frac{1}{2}\theta[M^{12}-M^{21}]} = e^{\theta M^{21}} \quad (1.6)$$

Now we recall concepts from the lectures. A vector representation:

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} \quad (1.7)$$

Theorem 1.1.1 Vector Representation of the Lorentz Algebra.

$$(M^{\rho\sigma})^{\mu}_{\nu} = \eta^{\rho\mu}\delta^{\sigma}_{\nu} - \eta^{\sigma\mu}\delta^{\rho}_{\nu} \quad (1.8)$$

Proof.

For (1.8) to be a Representation of the Lorentz Algebra it needs to satisfy (1.1) ($M \leftrightarrow S$).

$$\begin{aligned} \left[(M^{\rho\sigma}), (M^{\alpha\beta}) \right]^{\mu}_{\nu} &= (M^{\rho\sigma} M^{\alpha\beta})^{\mu}_{\nu} - (M^{\alpha\beta} M^{\rho\sigma})^{\mu}_{\nu} \\ &= (\eta^{\rho\mu}\delta^{\sigma}_{\lambda} - \eta^{\sigma\mu}\delta^{\rho}_{\lambda}) (\eta^{\alpha\lambda}\delta^{\beta}_{\nu} - \eta^{\beta\lambda}\delta^{\alpha}_{\nu}) - (\rho, \sigma) \leftrightarrow (\alpha, \beta) \\ &= \eta^{\rho\mu}\delta^{\sigma}_{\lambda}\eta^{\alpha\lambda}\delta^{\beta}_{\nu} - \eta^{\rho\mu}\delta^{\sigma}_{\lambda}\eta^{\beta\lambda}\delta^{\alpha}_{\nu} - \eta^{\sigma\mu}\delta^{\rho}_{\lambda}\eta^{\alpha\lambda}\delta^{\beta}_{\nu} + \eta^{\sigma\mu}\delta^{\rho}_{\lambda}\eta^{\beta\lambda}\delta^{\alpha}_{\nu} - (\rho, \sigma) \leftrightarrow (\alpha, \beta) \\ &= \eta^{\rho\mu}\eta^{\alpha\sigma}\delta^{\beta}_{\nu} - \eta^{\rho\mu}\eta^{\beta\sigma}\delta^{\alpha}_{\nu} - \eta^{\sigma\mu}\eta^{\alpha\rho}\delta^{\beta}_{\nu} + \eta^{\sigma\mu}\eta^{\beta\rho}\delta^{\alpha}_{\nu} - (\rho, \sigma) \leftrightarrow (\alpha, \beta) \\ &= \eta^{\rho\mu} (\eta^{\alpha\sigma}\delta^{\beta}_{\nu} - \eta^{\beta\sigma}\delta^{\alpha}_{\nu}) + \eta^{\sigma\mu} (\eta^{\beta\rho}\delta^{\alpha}_{\nu} - \eta^{\alpha\rho}\delta^{\beta}_{\nu}) - (\rho, \sigma) \leftrightarrow (\alpha, \beta) \\ &= \eta^{\rho\mu} (M^{\alpha\beta})^{\sigma}_{\nu} + \eta^{\sigma\mu} (M^{\beta\alpha})^{\rho}_{\nu} - \eta^{\alpha\mu} (M^{\rho\sigma})^{\beta}_{\nu} - \eta^{\beta\mu} (M^{\sigma\rho})^{\alpha}_{\nu}. \end{aligned}$$

which matches (1.1) □

Explicitly

$$M^{21} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (1.9)$$

We can consider the (1, 2) subspace of $\mathbb{R}^{1,3}$. In said subspace define, $B := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

So we have

$$\exp(\theta B) = \sum_{n=0}^{\infty} \frac{1}{n!} (-\theta)^n B^n.$$

Consider the parity of n , $k \in \mathbb{Z}^+$

$$\exp(\theta B) = \sum_{k=0}^{\infty} \frac{(-\theta)^{2k}}{(2k)!} B^{2k} + \frac{(-\theta)^{2k+1}}{(2k+1)!} B^{2k+1}.$$

Notice $B^{2k+1} = B \cdot B^{2k}$, $B^{2k} = (B^2)^k$. We have $B^2 = -\mathbf{1}_{2 \times 2}$ so $B^{2k} = (-1)^k \mathbf{1}_{2 \times 2}$ and $B^{2k+1} = (-1)^k B$. Thus

$$\exp(\theta B) = \sum_{k=0}^{\infty} \frac{(-1)^{2k} (\theta)^{2k}}{(2k)!} (-1)^k \mathbf{1}_{2 \times 2} + \frac{(-1)^{2k+1} (\theta)^{2k+1}}{(2k+1)!} (-1)^k B \quad (1.10)$$

$$\begin{aligned} \exp(\theta B) &= \sum_{k=0}^{\infty} \frac{(-1)^{3k} (\theta)^{2k}}{(2k)!} \mathbf{1}_{2 \times 2} + \frac{(-1)^{3k+1} (\theta)^{2k+1}}{(2k+1)!} B \\ \exp(\theta B) &= \sum_{k=0}^{\infty} \frac{(-1)^k (\theta)^{2k}}{(2k)!} \mathbf{1}_{2 \times 2} - \frac{(-1)^k (\theta)^{2k+1}}{(2k+1)!} B \end{aligned}$$

$$\exp(\theta B) = \cos \theta \mathbf{1}_{2 \times 2} - \sin \theta B.$$

This gives us the following

$$\exp(\theta B) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \quad (1.11)$$

Now we consider the $(0, 3)$ subspace we have $A := \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ then

$$\exp(\theta A) = \exp(\theta \mathbf{0}_{2 \times 2}) = \mathbf{1}_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (1.12)$$

Combining the subspaces we get

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We notice that we have a familiar rotation matrix around the x_3 axis. Since we have that $\sin \theta = \sin \theta + 2\pi$ and $\cos \theta = \cos \theta + 2\pi$ we see that a vector under this rotation will return to its original position after 2π rotations.

1.2 Part (B)

$$S[\Lambda] = e^{\frac{i}{2} \Omega_{\mu\nu} S^{\mu\nu}} = e^{\theta S^{21}} = e^{\frac{\theta}{4} (\gamma^2 \gamma^1 - \gamma^1 \gamma^2)} \quad (1.13)$$

The gamma matrices have the representation:

$$\gamma^0 = \begin{bmatrix} \mathbf{1}_{2 \times 2} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & -\mathbf{1}_{2 \times 2} \end{bmatrix} \quad \gamma^i = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \sigma^i \\ -\sigma^i & \mathbf{0}_{2 \times 2} \end{bmatrix} \quad (1.14)$$

Likewise, the Pauli matrices:

$$\sigma^1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma^2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \sigma^3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (1.15)$$

We have $\sigma^1\sigma^2 = i\sigma^3 = -\sigma^2\sigma^1$. So $[\gamma^1, \gamma^2] = -2i \begin{bmatrix} \sigma^3 & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \sigma^3 \end{bmatrix}$ We once again consider subspaces:

(x_0, x_2) corresponds to $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and (x_1, x_3) corresponds to $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. Since A is the identity, taken to any power it is still the identity. Likewise for B it is negative the identity so taken to some power m it returns $(-1)^m$ times the identity.

$$\begin{aligned} \exp\left(-\frac{\theta}{2}A\right) &= \sum_{n=0}^{\infty} \frac{(-1)^n \left(-\frac{i\theta}{2}A\right)^n}{n!} \\ \exp\left(-\frac{\theta}{2}A\right) &= \sum_{n=0}^{\infty} \frac{(-1)^n \left(-\frac{i\theta}{2}\right)^n}{n!} (A)^n \\ \exp\left(-\frac{\theta}{2}A\right) &= e^{-i\frac{\theta}{2}} \mathbf{1}_{2 \times 2}. \end{aligned}$$

And we have,

$$\begin{aligned} \exp\left(-\frac{\theta}{2}B\right) &= \sum_{n=0}^{\infty} \frac{(-1)^n \left(-\frac{i\theta}{2}B\right)^n}{n!} \\ \exp\left(-\frac{\theta}{2}B\right) &= \sum_{n=0}^{\infty} \frac{(-1)^n \left(-\frac{i\theta}{2}\right)^n}{n!} (B)^n \\ \exp\left(-\frac{\theta}{2}B\right) &= e^{i\frac{\theta}{2}} \mathbf{1}_{2 \times 2} \end{aligned}$$

Thus explicitly we have

$$S[\Lambda(\theta)] = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{-i\frac{\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{i\frac{\theta}{2}} \end{bmatrix} \quad (1.16)$$

Consider $\theta \rightarrow \theta + 2\pi$. $e^{-i\pi} = e^{i\pi} = -1$.

$$S[\Lambda(\theta + 2\pi)] = - \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 & 0 & 0 \\ 0 & e^{i\frac{\theta}{2}} & 0 & 0 \\ 0 & 0 & e^{-i\frac{\theta}{2}} & 0 \\ 0 & 0 & 0 & e^{i\frac{\theta}{2}} \end{bmatrix} \quad (1.17)$$

We see that we get the famous negative sign for the spinor representation of the Lorentz group. We require a rotation of 4π to get back to the original matrix. We have the double XXX

Question IV

We are given the Lagrangian $\mathcal{L}(\phi, \partial_\mu \phi, \psi, \bar{\psi})$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \bar{\psi} (i\gamma^\mu \partial_\mu - M) \psi - ig \bar{\psi} \gamma^5 \psi \phi \quad (1.18)$$

we want to show it is symmetric under the transformation

$$\phi(t, \vec{x}) \rightarrow -\phi(t, -\vec{x}), \quad \psi(t, \vec{x}) \rightarrow \gamma^0 \psi(t, -\vec{x}) \quad (1.19)$$

First we may make note of some convenient objects in the $(+, -, -, -)$ convention.

1. $\bar{\psi} \equiv \psi^\dagger \gamma^0$
2. $\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \mathbf{1}$ $\mu = \nu = 0$ gives $2(\gamma^0)^2 = -2$ so $(\gamma^0)^2 = 1$.
3. $\bar{\gamma}^0 = \gamma^0 (\gamma^0)^\dagger \gamma^0 = (\gamma^0)^2 \gamma^0 = \gamma^0$.

$$\begin{aligned} \mathcal{L} \rightarrow \mathcal{L}' &= \frac{1}{2} (-\partial_\mu \phi) (-\partial^\mu \phi) - \frac{1}{2} m^2 (-\phi) (-\phi) + \overline{(\gamma^0 \psi)} (i\gamma^\mu \partial_\mu - M) (\gamma^0 \psi) - ig \overline{(\gamma^0 \psi)} \gamma^5 (\gamma^0 \psi) (-\phi) \\ &= \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \psi^\dagger \gamma^0 \overline{(\gamma^0)} (i\gamma^\mu \partial_\mu - M) (\gamma^0 \psi) - ig \psi^\dagger \gamma^0 \overline{(\gamma^0)} \gamma^5 (\gamma^0 \psi) (-\phi) \\ &= \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \psi^\dagger \gamma^0 \gamma^0 (i\gamma^\mu \partial_\mu - M) (\gamma^0 \psi) - ig \psi^\dagger \gamma^0 \gamma^0 \gamma^5 (\gamma^0 \psi) (-\phi) \\ &= \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \psi^\dagger (i\gamma^\mu \partial_\mu - M) (\gamma^0 \psi) - ig \psi^\dagger \gamma^5 (\gamma^0 \psi) (-\phi). \end{aligned}$$

where we used XXX

Consider the following $(j = 1, 2, 3)$. We have $\gamma^\mu \gamma^\nu = \gamma^\nu \gamma^\mu \iff \mu = \nu$, otherwise $\gamma^\mu \gamma^\nu = -\gamma^\nu \gamma^\mu$. Note $j \neq 0$.

$$\begin{aligned} \psi^\dagger (i\gamma^\mu \partial_\mu - M) (\gamma^0 \psi) &= \psi^\dagger i\gamma^\mu \partial_\mu \gamma^0 \psi - \psi^\dagger M \gamma^0 \psi \\ &= \psi^\dagger i\gamma^0 \partial_0 \gamma^0 \psi - \psi^\dagger i\gamma^j \partial_j \gamma^0 \psi - \gamma^0 \psi^\dagger M \psi \\ &= \gamma^0 \psi^\dagger i\gamma^0 \partial_0 \psi + \gamma^0 \psi^\dagger i\gamma^j \partial_j \psi - \bar{\psi} M \psi \\ &= \bar{\psi} i\gamma^0 \partial_0 \psi + \bar{\psi} i\gamma^j \partial_j \psi - \bar{\psi} M \psi \\ &= \bar{\psi} (i\gamma^\mu \partial_\mu - M) \psi. \end{aligned}$$

We also have that $(\gamma^0)^\dagger = \gamma^0$

$\gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3$. Thus $\gamma^5 \gamma^0 = (i\gamma^0 \gamma^1 \gamma^2 \gamma^3) \gamma^0 = (-1)^3 \gamma^0 \gamma^5 = -\gamma^0 \gamma^5$.

$$\begin{aligned} -ig \psi^\dagger \gamma^5 (\gamma^0 \psi) (-\phi) &= +ig \gamma^0 \psi^\dagger \gamma^5 \psi (-\phi) \\ &= -ig \bar{\psi} \gamma^5 \psi \phi. \end{aligned}$$

Therefore the final form of the Lagrangian is

$$\mathcal{L}' = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \bar{\psi} (i\gamma^\mu \partial_\mu - M) \psi - ig \bar{\psi} \gamma^5 \psi \phi = \mathcal{L}.$$

Thus the Lagrangian is invariant under the transformation.